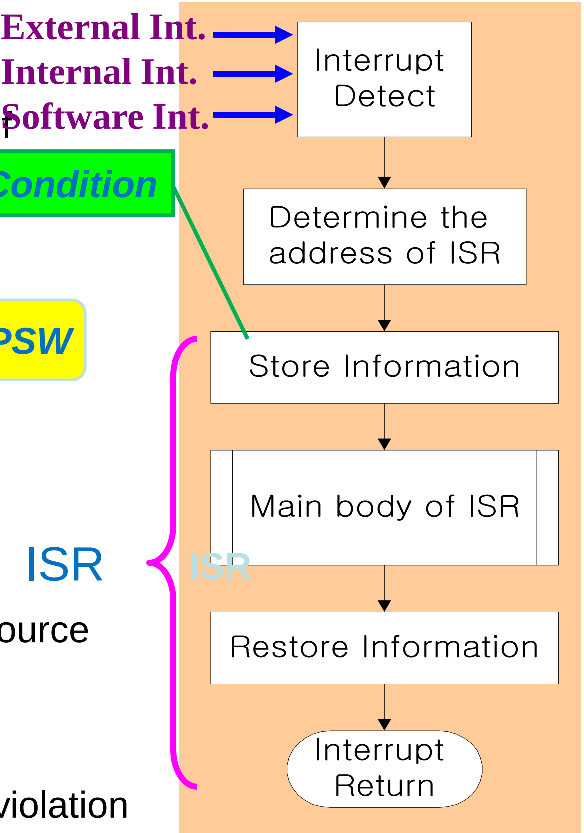
- Conditional Branch : *Tab. 8-11*
- Subroutine Call and Return
  - CALL : $SP \leftarrow SP - 1$: Decrement stack point
    $M[SP] \leftarrow PC$: Push content of PC onto the stack
    $PC \leftarrow Effective\ Address$: Transfer control to the subroutine
  - RETURN : $PC \leftarrow M[SP]$: Pop stack and transfer to PC
    $SP \leftarrow SP + 1$: Increment stack pointer
- Program Interrupt
  - Program Interrupt
    - Transfer program control from a currently running program to another service program as a result of an external or internal generated request
    - Control returns to the original program after the service program is executed
  - Interrupt Service Program Subroutine Call
    - 1) An interrupt is initiated by an internal or external signal (*except for software interrupt*)
      » A subroutine call is initiated from the execution of an instruction (CALL)
    - 2) The address of the interrupt service program is determined by the hardware
      » The address of the subroutine call is determined from the address field of an instruction
    - 3) An interrupt procedure stores all the information necessary to define the state of the CPU
      » A subroutine call stores only the program counter (*Return address*)

- Program Status Word (PSW)
  - The collection of all status bit conditions in the CPU
- Two CPU Operating Modes
  - Supervisor (*System*) Mode : Privileged Instruction
    - » When the CPU is executing a program that is part of the operating system
  - User Mode : User program
    - » When the CPU is executing an user program

PC, CPU Register, Status Condition

CPU operating mode is determined from special bits in the PSW



- Types of Interrupts
  - 1) External Interrupts
    - come from I/O device, from a timing device, from a circuit monitoring the power supply, or from any other external source
  - 2) Internal Interrupts or TRAP
    - caused by register overflow, attempt to divide by zero, an invalid operation code, stack overflow, and protection violation
  - 3) Software Interrupts
    - initiated by executing an instruction
    - used by the programmer to initiate an interrupt procedure at any desired point in the program

- 8-8 Reduced Instruction Set Computer (RISC)
  - Complex Instruction Set Computer (CISC)
    - Major characteristics of a CISC architecture
      - 1) A large number of instructions - typically from 100 to 250 instruction
      - 2) Some instructions that perform specialized tasks and are used infrequently
      - 3) A large variety of addressing modes - typically from 5 to 20 different modes
      - 4) Variable-length instruction formats
      - 5) Instructions that manipulate operands in memory
  - Reduced Instruction Set Computer (RISC)
    - Major characteristics of a RISC architecture
      - 1) Relatively few instructions
      - 2) Relatively few addressing modes
      - 3) Memory access limited to load and store instruction
      - 4) All operations done within the registers of the CPU
      - 5) Fixed-length, easily decoded instruction format
      - 6) Single-cycle instruction execution
      - 7) Hardwired rather than microprogrammed control